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How autonomous research shapes scientific understanding

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Figure 1 is an ImageGen concept. All quantitative figures use retained data or original native quantitative artwork.

Autonomous science promises to translate experimental agency into advances in scientific understanding. Yet how agents organize inquiry and turn experimental experience into knowledge remains poorly understood. Here we introduce ChemWorld, programmable chemical worlds combining autonomous experimentation with independent control of underlying laws and supplied information. Across chemical systems, model configurations and repeated investigations, we connect research objectives and prior knowledge to experimental choices, evolving explanations and subsequent judgments. Agents propose and test explanatory possibilities, establish useful local relationships and retract unsupported expectations. Similar predictive accuracy can nevertheless rest on different evidence histories, while successful local research can leave mechanisms and wider applications unresolved. Matched chemical worlds distinguish the information acquired by an experiment from the interpretation a researcher gives it; research-opportunity comparisons examine the consequences of conducting inquiry. Together, these studies make autonomous research itself an object of empirical investigation, linking how exploration is organized to how scientific understanding forms, develops and guides further inquiry.

Scientific discovery involves more than deriving answers from established premises: it also introduces ways of understanding phenomena that open new directions for inquiry. Einstein began his 1905 paper on special relativity with a tension between the observable effects of magnet–conductor relative motion and the asymmetry of their prevailing theoretical descriptions. He introduced principles from which consequences for simultaneity, time and motion could be

derived[1]. This episode illustrates the creative work of reconsidering the premises through which the world is understood.

Abduction concerns a central part of this activity: proposing explanatory hypotheses worth investigating. In the Peircean account, abduction introduces possible explanations, deduction develops their testable consequences, and empirical testing and induction assess what should be retained, revised or abandoned[2]. An explanation can thus be both a product of research and a starting point for further exploration. Experiments evaluate existing judgments while also changing which phenomena demand explanation and which questions become worth pursuing.

An autonomous researcher does more than act on the world. By choosing experiments and measurements, it also shapes the experience from which its scientific judgments form: which phenomena become visible, which explanations encounter a test and which results guide the next investigation. As AI scientists and self-driving laboratories connect increasingly elaborate experimental cycles, this organization of inquiry becomes part of the scientific activity delegated to machines[3,4]. The promise of autonomous science therefore concerns both the expansion of experimental capability and the transformation of experimental experience into scientific understanding.

Robot scientists and self-driving laboratories have demonstrated hypothesis testing, materials discovery and reaction optimization[5–8]. Language-model agents extend the connection between scientific instructions, tools and laboratory operations[9,10]. These advances make it possible to investigate a further question: how does an agent's own research change the relationships it uses to explain and predict the world? A successful procedure establishes an experimental achievement, and a correct forecast establishes predictive accuracy. Neither alone reveals the evidence from which a judgment arose or the scope that evidence supports[11,12].

The difficulty is that the evidence available to an autonomous researcher is itself an outcome of its research. An initial explanation can suggest a productive experiment; the observation can support, narrow or displace that explanation; the resulting judgment can then guide another intervention. A sequence of successful experiments may establish a useful local relationship without resolving how it extends beyond the observed conditions. Conversely, an unsuccessful intervention can advance understanding by ruling out an expectation. Studying this interaction

requires connecting what the researcher initially had reason to expect, what its experiments actually tested and what it subsequently treated as established[13–15].

We study these connections in ChemWorld. Our central question is how AI organizes inquiry to develop scientific understanding when its initial information does not determine the answer, and how developing understanding in turn shapes exploration. Programmable laws and independently specified public information allow research histories and their consequences to be compared. Cross-system studies establish the breadth of operating and predictive outcomes; a five-model comparison identifies different routes to evidence and extrapolation; repeated investigations locate changes in working judgments; and a matched research-opportunity comparison tests the contribution of conducting research. Additional matched chemical worlds connect the physical information in selected experiments to the explanations proposed and revised. Together, these studies examine the organization of inquiry, the formation of knowledge and their continuing interaction.

Making autonomous scientific inquiry experimentally tractable

ChemWorld combines materials, operations and instruments into stateful virtual chemical systems governed by thermodynamic, kinetic and coupled process relationships. Reaction, thermal processing, phase equilibrium, separation and crystallization components can be connected into multistage procedures. Sample states persist, so an upstream reaction or thermal history can change a later separation or measurement. Agents choose quantities, processing sequences, measurements, repetitions and subsequent experiments within a fixed resource allowance. Their choices determine both the procedures they develop and the observations from which they reason. An evaluator independently controls world parameters and public information, records committed operations and obtains withheld reference outcomes (Fig. 1). The agent does not receive those outcomes during research or assessment.

This arrangement makes the agent's research process experimentally accessible. Knowledge tests and supplied-data tasks assess important scientific abilities[16,17], and interactive discovery environments already connect action to observation[18–21]. ChemWorld connects an agent's evidence selection to evolving public judgments and their later use in composable chemical pro-

cesses. An attempted operation can be rejected before changing the sample; an observation can support a relationship locally without establishing it under every compatible intervention. Execution records and independently evaluated world responses allow these distinctions to be checked.

The broad study comprises 240 Sol campaigns across electrochemistry, partitioning, reaction processing, aqueous equilibrium, reaction-coupled crystallization and purification, plus 60 equilibrium campaigns from four other model configurations. The systems support different scientific questions: developing an effective electrical or thermal procedure, characterizing a concentration response, or recovering material while respecting quality constraints. Discovery and optimization objectives, partial prior information and experimental budgets provide complementary comparisons within the specified study blocks. Each block uses five fixed worlds and three information conditions: Opaque retains the operational contract without an additional instance dossier; Aligned supplies correct but incomplete information; MisIndexed supplies a specified mismatch. These manipulations change starting conditions for inquiry while preserving the corresponding experimental interface.

The current process analysis adds temporal resolution through 45 equilibrium sessions and 45 crystallization sessions conducted under a corrected, consistent recovery definition. Each system covers five fixed worlds, three information conditions and three fresh model sessions per condition. These 90 sessions contain 80 complete research-and-assessment chains and 83 valid forecast stages; incomplete sessions remain in the denominator. Earlier crystallization sessions affected by a recovery-description mismatch are retained separately. The broad study establishes cross-system contrasts, whereas the repeated studies examine how particular judgments arise and vary across research histories; their protocols and denominators are not pooled.

Contemporaneous notes connect publicly stated judgments to the observations available at that time. After research, agents seal a scientific account, predict outcomes under twelve new interventions without laboratory access, and reflect without receiving reference truth. Structured process analysis identifies 92 observable judgment–experiment–revision windows across 65 of the current 90 sessions (Extended Data Fig. 2). The other 25 sessions lack a complete window under this recording definition; this does not establish an absence of scientific learning. A sep-

arate contemporaneous comparison assigns 30 fresh sessions to research or no research under a common assessment protocol. Across these layers, we distinguish a relationship's agreement with world outcomes, the evidence available to support it and the way the researcher subsequently uses it.

A further experiment separates the information acquired through exploration from the interpretation given to it. Five matched pairs of aqueous worlds share public framing and selected parameters, but differ in whether the precipitation network includes an additional neutral aqueous ion pair. Two supplied ultradilute assays per world have closely matched underlying responses; autonomous loading, dilution and scale choices can reveal larger differences. Three fresh sessions per world yield thirty complete investigations, 360 final assays and eighty notebook versions. Replaying each chosen action sequence in its paired world shows that all thirty sessions acquire at least one observation differing by three assay standard deviations in two response channels (Fig. 4a). This establishes available physical information, not successful identification of a mechanism. The task supplies the research domain and possible experimental dimensions; the questions developed within it are not inventions without prior framing.

The self-organization and adaptation of scientific inquiry

Research starts from goals, available information and a constrained action space. The historical electrochemical and reaction studies ask agents either to develop and test scientific understanding or to optimize a public operating score. These commissions lead to genuine experimental achievements, but operating and predictive gains are not interchangeable (Fig. 2a,b). Optimization produces the better retested electrochemical procedure in 26 of 30 matched comparisons; among those 26, prediction error is lower in thirteen and higher in thirteen. In reaction processing, discovery produces both the better procedure and more accurate score predictions in seventeen of thirty pairs. The relation between a research objective and its scientific consequences therefore depends on what the inquiry actually establishes.

One electrochemical pair makes the distinction concrete (Fig. 2c,d). Discovery compares materials and conditions before increasing duration and current cap; optimization screens material combinations and then refines operating conditions for a promising pair. The optimization rec-

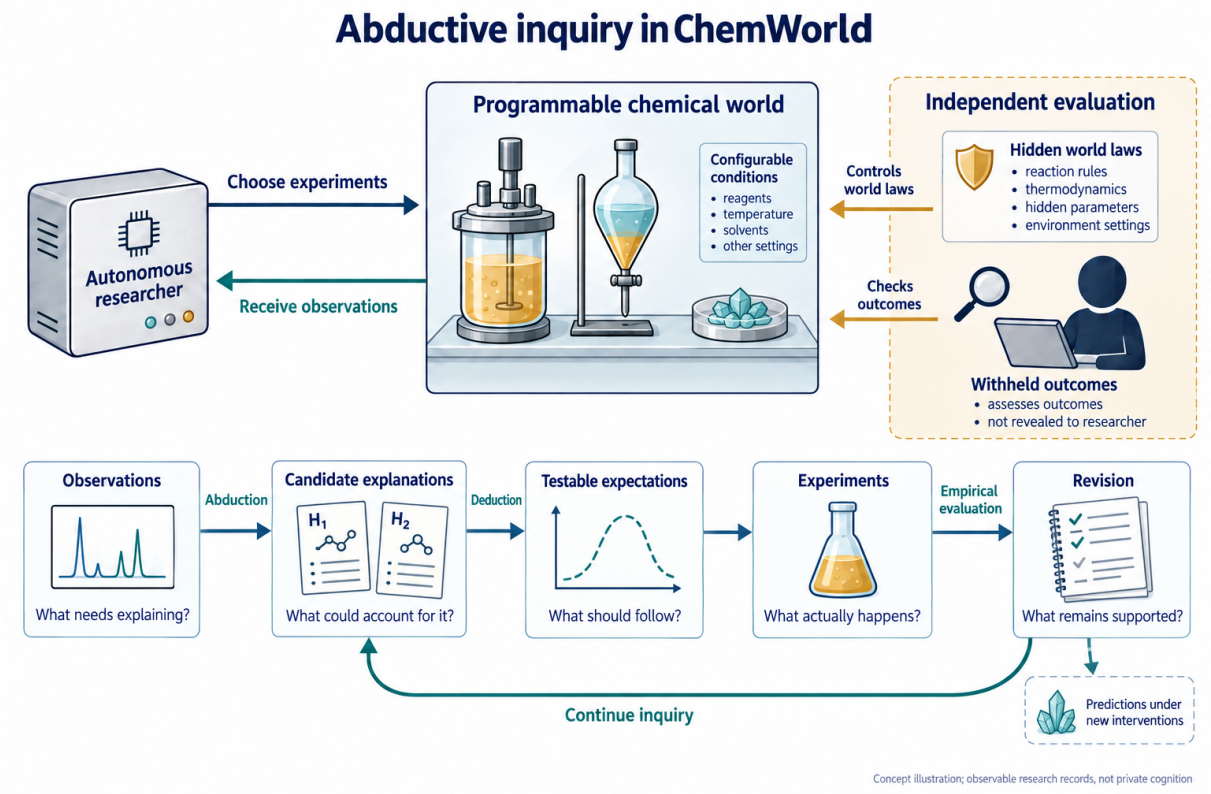


Figure 1 | Abductive inquiry as an observable research process. ImageGen concept illustration. An autonomous researcher chooses interventions and receives public observations from a persistent chemical world. The evaluator separately controls hidden laws and checks withheld outcomes. The lower cycle distinguishes proposing candidate explanations, deriving testable expectations, performing experiments and revising what remains supported. Revised judgments can guide further inquiry and predictions under new interventions. This is an analytical framework for observable research records, not a claim that every action completes the cycle or a direct measurement of private cognition.

ommendation retests at 0.756 against discovery's 0.718, but its prediction mean absolute error (MAE) is 0.229 against 0.088. Its later prediction rationale describes several poorly performing cross-pair combinations as low-activity. That account accompanies a forecast of 0.070 against a reference of 0.494 for one combination under new conditions, while another forecast remains accurate at 0.570 against 0.572. The histories expose what a single endpoint obscures: success in developing a procedure can coexist with uneven judgments about the broader experimental space.

Prior information provides another entry into the organization of inquiry. In corrected crystallization, all fifteen aligned-information sessions start with solvent S2 and catalyst C1, whereas all fifteen sessions without an instance dossier start with S0 and C0. Under the specified mismatched dossier, thirteen start with S2 and fourteen with C1. Mean committed solvent coverage

is 2.93, 4.00 and 2.40 for Aligned, Opaque and MisIndexed, respectively. These descriptive differences locate how supplied information enters material selection; wider coverage alone does not establish better learning. In reaction processing, partial parameter knowledge has a clear predictive benefit: Aligned reduces macro MAE from 0.102 to 0.057 under discovery and from 0.139 to 0.084 under optimization, improving on Opaque in four of five worlds under each commission.

Research paths also differ under the same objective and scientific conditions. In the five-model equilibrium comparison, researchers can choose loading, dilution and addition order before predicting pH, dissociation and precipitation. The hidden laws produce a near-constant dissociation fraction over the higher-concentration region, with a changed response at extreme dilution. Eleven of Astra's fifteen campaigns assay positively loaded samples at nominal concentrations no greater than 1 mM, compared with none of fifteen for each of Sol, GPT-5.5, Luna and Terra (Fig. 3a). This difference describes which evidence these particular campaigns acquire, rather than a fixed exploratory trait of a model.

An Astra campaign illustrates the scientific use of newly acquired evidence (Fig. 3d). Its first three assays show dissociation of 6.3–7.4%; batch five tests 18.5 micromolar and observes 53.8%. The later scientific account distinguishes the high-concentration plateau from the dilute response, and the sealed forecast at 13.3 micromolar is 58.80% with an 80% interval of 54.20–63.40%, against a reference mean of 57.97%. Sol/Opaque takes a different route: despite remaining outside the dilute regime during research, its five-world dilute macro MAE is 0.01833. Accurate prediction can therefore accompany either a newly observed change or extrapolation beyond local observations. Direct sampling of every target condition is not a prerequisite for successful inference.

These cross-system and model comparisons establish variation in the evidence histories that precede a judgment. Temporal records then allow us to ask how feedback enters the next research choice. Some campaigns continue a scan planned at the outset; others repeat a borderline recipe, narrow a parameter interval or formulate a new test. Counts of explored materials cannot distinguish these activities. Historical electrochemical discovery and optimization campaigns, for

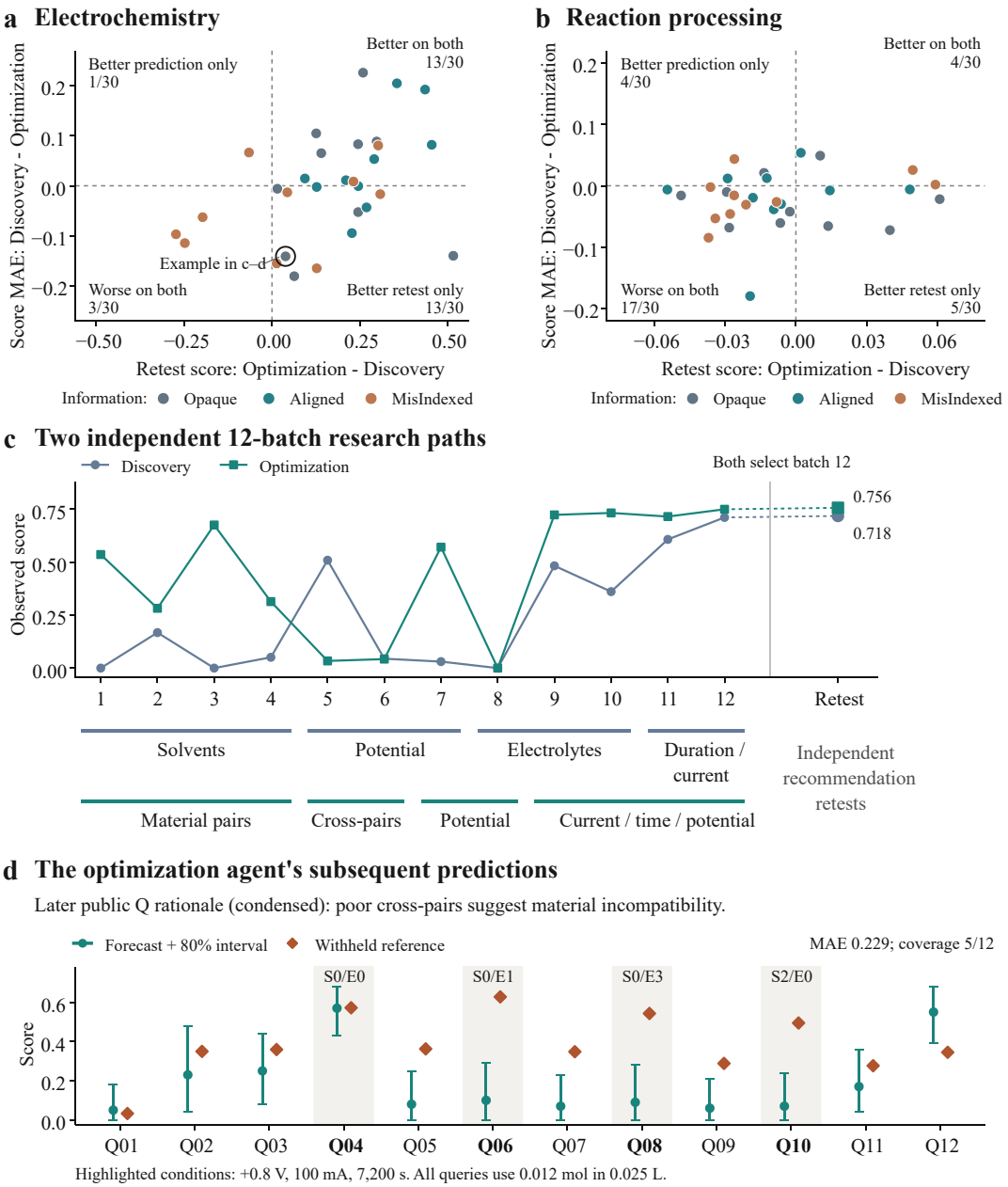


Figure 2 | Experimental achievement and predictive accuracy across research objectives. a,b, Thirty matched discovery–optimization pairs in electrochemistry and reaction processing. Positive horizontal values favour optimization on retest score; positive vertical values favour optimization on prediction MAE. Quadrant labels refer to optimization relative to discovery. c, Two independent twelve-batch electrochemical histories and their recommendation retests; batch-phase labels summarize executed choices. d, All twelve optimization forecasts from that pair, with original 80% intervals and withheld reference outcomes. The later prediction rationale is condensed, not a contemporaneous hypothesis record. The figure retains historical protocols and is not pooled with the repeated process cohort.

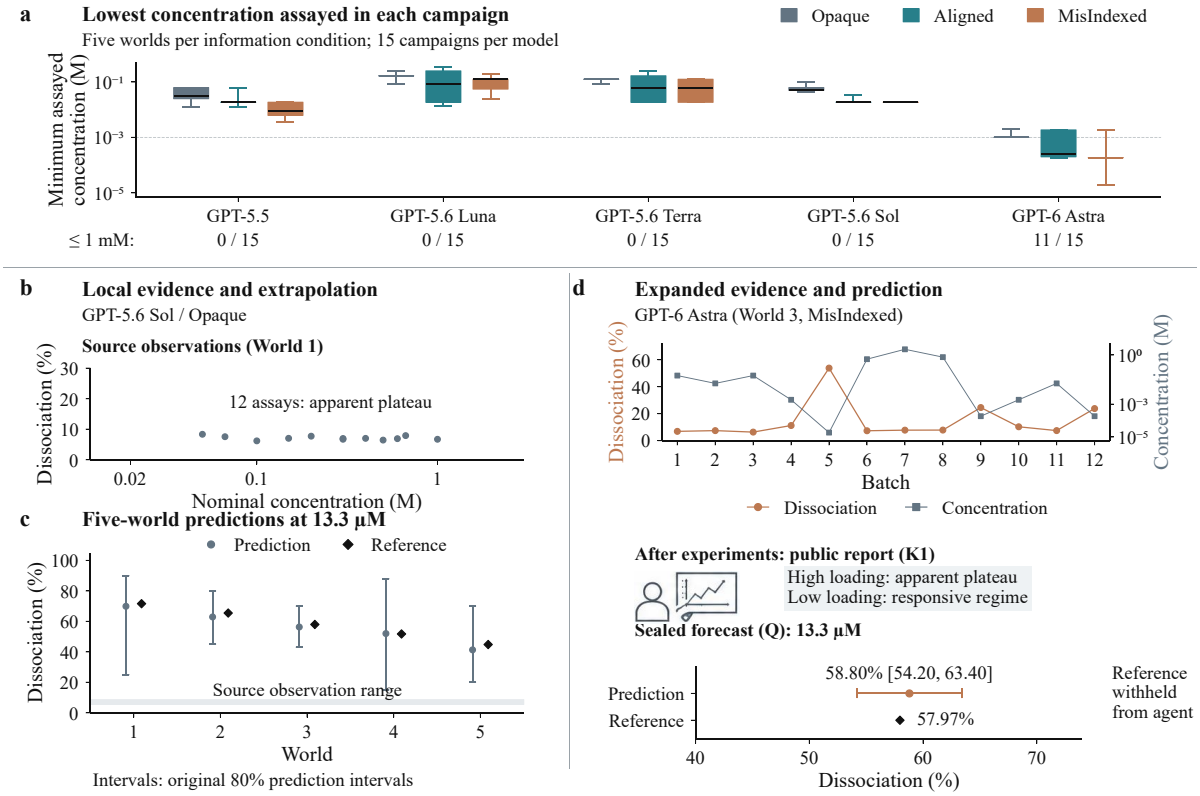


Figure 3 | Distinct routes from experimental evidence to prediction. a, Minimum positively loaded, assayed concentration for all 75 historical equilibrium campaigns. Each information-arm box contains five world values: median, first and third quartiles, and minimum–maximum whiskers. Quartiles are calculated before logarithmic display. The 1 mM threshold is descriptive. b, Twelve source assays in one Sol/Opaque campaign. c, The most dilute forecast in all five Sol/Opaque worlds, with original 80% intervals; the shaded band marks source dissociation values. d, All twelve batches of one Astra/MisIndexed campaign, its condensed post-experiment account and a sealed forecast. The account is retrospective, not a quotation of contemporaneous reasoning. These data distinguish expanded evidence acquisition from extrapolation beyond local evidence, without assigning either route a model-wide cognitive mechanism.

example, cover similar numbers of solvents and both cover all four at the larger budget, despite their differing operating and predictive outcomes.

The matched-world investigations make the content of such adaptation explicit (Fig. 4). One researcher proposes that a sharp precipitation onset with a buffered dissolved response would support a solubility-limited interpretation, whereas smooth changes would also admit an empirical partition description. After six assays, it proposes denser sampling of the concentration response, a central repeat and an increasing-loading step. The next six batches implement these investigations. The final account retains a smooth, monotone relationship, while leaving a broadened threshold and an empirical partition description unresolved. Its forecast later states: “Precipitation estimates use a smooth monotone concentration response because no sharp threshold was

resolved.” The research develops a usable relationship and delimits competing interpretations without uniquely identifying the underlying network.

A second trajectory tests a more specific consequence. Continued acidification weakens a strict plateau expectation, and the researcher proposes diluting a concentrated sample to distinguish concentration dependence from persistence of the high-loading state. The next batch dilutes nominal loading from 0.75 to 0.375 M: normalized pH rises from 0.08045 to 0.10803 and the final precipitation signal is 0.08026, compared with 0.10004 measured before dilution. The account supports approximate reversibility while retaining uncertainty about small hysteresis; its nearest separately prepared comparator is 0.4 M, not an exact concentration match. The explanatory value lies in a stated alternative becoming a prospective physical test, followed by a qualified judgment about what that test establishes.

Across the corrected crystallization cohort, 52 of 58 selected immediate plans match the ensuing committed actions and six match partly. This measures a public plan–action connection within observable windows, not a discovery rate or an effect of writing the note. Together with the continuous equilibrium case, these records make the adaptation of inquiry observable: an experimental history can enter a revised expectation and a subsequent action. What those actions establish requires examining the content and scope of the resulting relationships.

The formation and evolution of scientific understanding through experimentation

Scientific relationships acquire practical meaning when they support a further test, constrain an explanation or guide a new intervention (Fig. 5). In a corrected crystallization study, a near-threshold fines result leads to a prospective proposal to lower the reagent charge from 0.03 to 0.02 mol. The next batch implements that proposal and reduces fines from 0.529 to 0.387, crossing the 0.50 threshold. The result supports a local recipe improvement. Because the absolute catalyst amount remains fixed, however, the catalyst-to-reagent ratio also changes: the experiment does not separately identify a concentration mechanism. What is established is a useful relation between the tested recipes and their outcomes, with a narrower scope than a uniquely identified causal law.

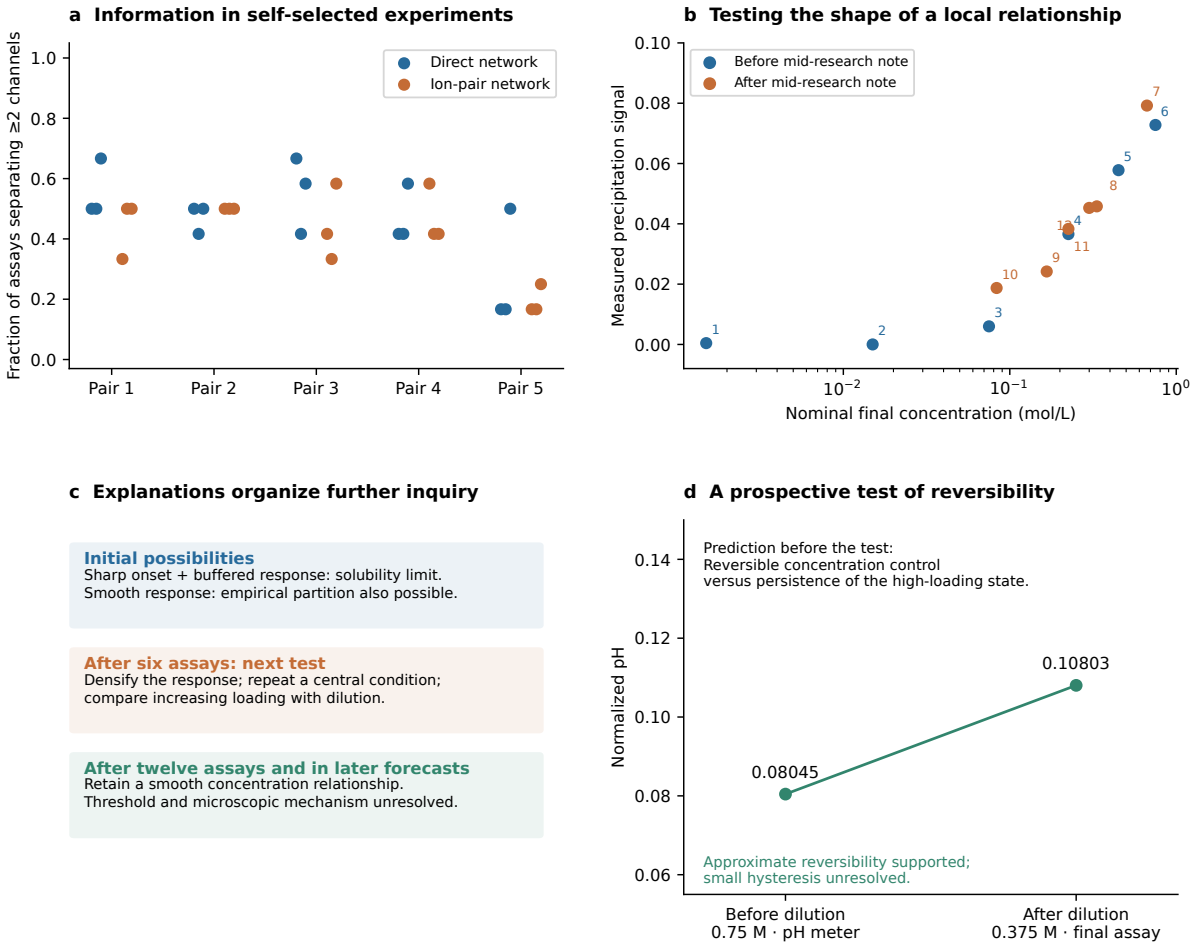


Figure 4 | From explanatory possibilities to experiments and bounded relationships. a, Fraction of the twelve chosen final assays separating the two matched physical networks by at least three assay standard deviations in two or more channels. Each point is one session, not a mechanism-identification score. b, Twelve measured precipitation responses in one session, before and after the mid-research note; labels identify batch order. The later six batches include denser sampling, a central repeat and an increasing-loading step. c, Public expectation, planned test and subsequent interpretation in that same session, summarized from the notes and sealed forecast. d, A second session's staged dilution and its measured pH response, with the conditional test stated before execution. The first reading is from the pH meter and the second from the final assay; they are not replicates. Concentrations are nominal added amounts divided by added solvent, and points are single observations. Mechanisms remain unresolved even where empirical relationships become useful.

A second crystallization study tests an interpolation. After observing different recovery and fines outcomes at nearby solvent volumes, the researcher proposes that 0.065 L should preserve the recovery benefit while improving fines. The next experiment instead increases fines from 0.512 to 0.601, and the final note retracts the expectation. An unsuccessful operating intervention therefore produces an explicit, evidence-responsive revision. Together with the successful lower-charge test, it shows two ways experimentation can advance a working judgment: establishing a local improvement or withdrawing a prediction the observation does not support.

Whether an intervention actually occurred also determines what can be learned from it. In seed case B, a researcher plans approximately 1 g of cumulative seed addition, but only 0.05 g commits and nineteen subsequent requests are rejected. Its later note correctly recognizes that the high-dose test remains unexecuted. Seed case C encounters analogous rejected requests yet describes 1 g as tested and uses the outcome against a seed-based explanation. These contrasting accounts concern the evidential meaning assigned to the same kind of execution limitation. Additional thermal-history and near-threshold repetition cases remain in the supplementary record.

The new matched-world records expose another transition: observations acquire explanatory significance through the researcher's reconstruction of its own experiments (Extended Data Fig. 6). In one session, a rejected solvent addition is later counted as completed, turning an actual concentration near 0.025 M into a reported 0.0083 M. Comparing this purportedly dilute sample with a 0.0125 M assay creates an apparent reversal in the pH trend. A path- or geometry-dependent explanation first appears in the terminal note, then enters the scientific account and subsequent forecast rationales; its proposed follow-up test is not executed. The observations are real, but the experimental comparison offered as their explanation rests on an incorrect history.

A contrasting session revises such a reconstruction. Following a failed termination and rejected solvent addition, the researcher initially treats samples near 0.10 and 0.05 M as concentration matched and proposes a volume- or activity-dependent explanation for their pH difference. Later, after genuine matched-concentration experiments already included in its plan, the final note reconstructs the earlier sequence and explicitly withdraws that explanation. The corrected account carries into subsequent judgments. This is an observed withdrawal within autonomous inquiry, not an effect attributed to an imposed correction strategy. Together, these cases show that knowledge formation depends on both what the world reveals and what the researcher takes its own interventions to have tested.

An earlier research-opportunity trajectory, here called equilibrium study A, connects a valid local finding to its later use (Extended Data Fig. 5). Its initial weak-acid expectation is not supported by the near-constant dissociation observed in the first six assays. The next note proposes an alternative feed identity and batches eight to twelve switch the requested feed command. Execution semantics show that both commands add the same species: the intended identity con-

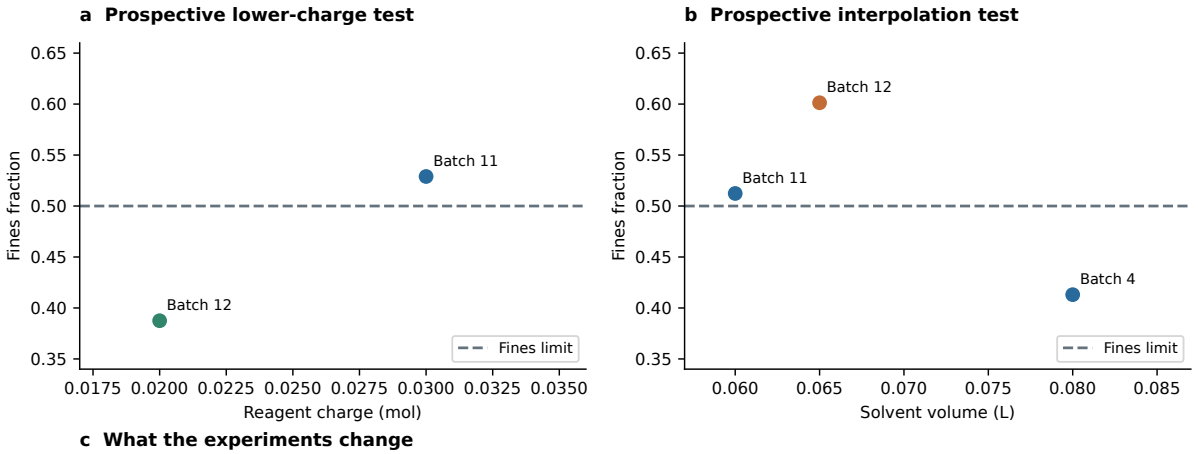
trast was not realized. The plateau remains supported locally, while the proposed mechanism remains unresolved. Nevertheless, when forecasting extreme dilution, the agent writes: “The point estimate still favors the experimentally supported fixed acidic background”. It predicts dissociation of 0.085 [0.035, 0.350], against a reference mean of 0.716, while acknowledging that this regime was untested. The matched no-research forecast is 0.569 [0.040, 0.980] and explicitly depends on an untested prior assumption. Its greater numerical accuracy does not establish a better experimental discovery.

That continuation is not inevitable. Two sessions in another fixed equilibrium world both observe plateaux at relatively high loading, yet their dilute macro errors are 0.16287 and 0.01462. For the same dilute query, they predict dissociation of 0.130 and 0.626 against a reference mean of 0.65463. The more accurate session uses a weak-acid continuation consistent with its plateau observations and explicitly describes it as conjectural. Together with the historical Sol/Opaque and Astra cases, these records distinguish several routes to a judgment: an observed change, a plausible extrapolation and a local relationship carried into an unresolved regime. Numerical correctness and experimental establishment remain separate questions.

The process evidence thus identifies scientifically consequential transitions rather than a single category of error. A failed experiment can narrow what the researcher has reason to expect; a successful intervention can establish a bounded improvement; and a correct recognition of local regularity can coexist with an unreliable application elsewhere. The point at which a judgment is formed and the point at which it is used need not have the same evidential status. Following both points explains why operational failure need not mean an absence of learning, and why a defensible local finding need not guarantee a defensible extrapolation.

The nature and conditions of scientific progress in autonomous inquiry

The preceding results locate how research is organized and how relationships acquire practical meaning. We next ask when those achievements carry into reliable judgments under new conditions. Prior knowledge, additional resources and the opportunity to conduct experiments can each contribute, but their benefits need not be uniform across the scientific questions a researcher encounters.



c What the experiments change

Judgment	Test intention	Executed evidence	Supported standing	Subsequent use
EQ component explanation	Change the acid source	Same species via two command names	Local plateau; mechanism unresolved	Background favoured in dilute forecast
Lower-charge expectation	Cross fines limit with useful recovery	Fines 0.529 to 0.387; ratio also changes	Local recipe improvement	Select tested batch 12
Interpolation expectation	Retain recovery; restore fines <0.50	Fines 0.512 to 0.601 at intermediate volume	Expectation retracted	Return to feasible batch 4

Figure 5 | Executed tests and changes in evidential standing. a, The two final lower-charge batches: fines cross the 0.50 limit while absolute catalyst amount remains fixed. b, The interpolation study's baseline and two later volumes: the intermediate volume does not achieve the predicted constraint crossing. Points are individual final assays; batch labels give chronology and no repeatability interval is inferred. c, Three histories connect the intended test, actual evidence, supported standing and subsequent use. The equilibrium row concerns a candidate explanation; the crystallization rows concern prospective empirical expectations. Terminal revisions are included even when no later source experiment occurs.

The equilibrium prior comparison contrasts with the positive reaction results. In the original Sol campaigns, correct partial information lowers macro MAE on nine queries from 0.01008 to 0.00465, yet increases it on the three most dilute queries from 0.01833 to 0.15862. Both directions occur in all five worlds. A separate repeated cohort shows the same aggregate directions with different magnitudes: 0.00907 to 0.00512 on nine queries and 0.09880 to 0.13205 on the dilute subset. These cohorts are reported separately. The model comparison further limits the interpretation (Extended Data Fig. 1): Luna and Terra improve with Aligned information in both regions, GPT-5.5 changes little, and Astra retains low dilute error across all three arms. The effect of partial knowledge is therefore conditional on both the researcher and the condition being judged; a local benefit does not determine its value elsewhere.

Research resources likewise support different kinds of progress. Historical independent twelve- and twenty-four-batch campaigns improve some electrochemical, partitioning and crystallization predictions without a corresponding increase in feasible crystallization recommendations or mean retested recovery. These are comparisons of complete resource envelopes, not matched continuations of a single research history. Combined with the goal contrasts in Fig. 2, they show why neither a more productive procedure nor a larger experimental budget alone specifies what scientific relationships have been learned.

The matched research-opportunity comparison asks a more direct question: does access to research change subsequent prediction under common initial scientific information (Fig. 6)? All thirty sessions use Opaque information: the public task, material identities, measurement scales, legal operations and initial state are available, but no instance-specific archival law or fitted relationship is supplied. Each of five equilibrium worlds receives three fresh sessions with twelve source batches and three without source experiments. Scientific inputs and sealed assessments are matched within each world and repeat. All thirty sessions terminate, all 180 planned source batches complete, and twenty-nine sessions provide valid forecasts. One research session loses its forecast and later stages to a network interruption; its observations and scientific account remain retained.

Research substantially improves prediction overall. Averaging valid sessions within each world and then weighting the five worlds equally, normalized macro mean absolute error falls from 0.132086 without research to 0.035632 with research, a 73.0% reduction. The direction holds in all five worlds. Yet for the three prospectively designated extreme-dilution queries, error rises from 0.061467 to 0.103677, a 68.7% increase, with deterioration in four of five worlds. Over the remaining nine queries it falls from 0.155625 to 0.012950. The comparison concerns the total opportunity to conduct research, including experiment selection, feedback and available recording tools; it does not isolate any one component.

Within this comparison, the same research opportunity improves overall prediction while worsening it in a defined region. All three response-specific point errors increase on the dilute subset, but by different amounts: Research-minus-NoResearch MAE is +0.00777 for normalized pH, +0.10795 for dissociation fraction and +0.01092 for precipitation signal. Dissociation accounts

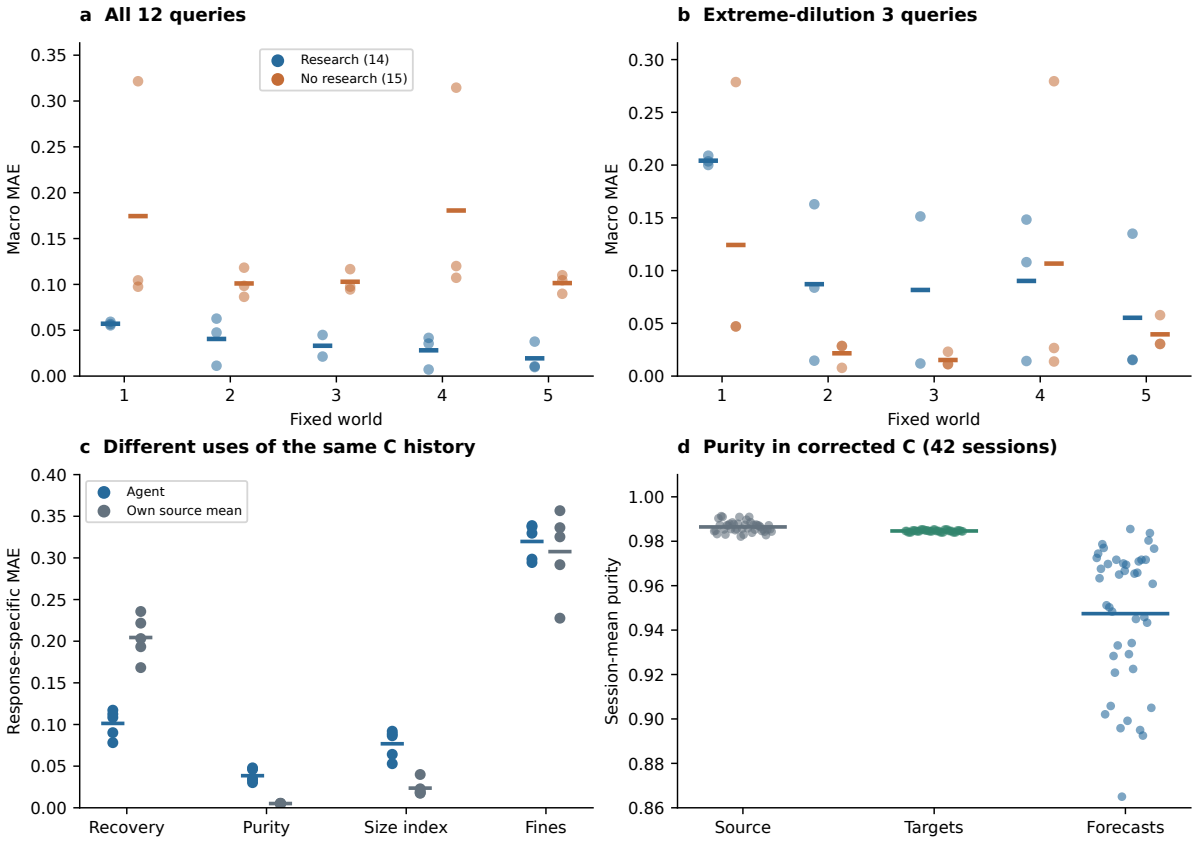


Figure 6 | Consequences of inquiry for subsequent judgment. a,b, Actual session macro errors for all twelve and the three extreme-dilution queries, respectively; short horizontal marks show within-world means. Fourteen Research and fifteen NoResearch forecasts are available; the missing Research forecast in world 3 is retained as missing. c, Corrected-crystallization forecast MAE versus each session's own source-mean reference. Each point is a world mean; bars average the five worlds equally. Responses retain their own public scales. d, Source, target and forecast purity means for the same 42 valid sessions; horizontal positions are deterministic offsets for visibility. These descriptive points and means are not confidence intervals.

for 85.2% of the macro-error increase. Figure 6 shows individual sessions and world means; the complete session-by-response and query-level results accompany the manuscript. Equilibrium study A supplies one observed route from a valid local finding to a distant judgment. It does not establish that this route explains every session-level difference or that acquiring more evidence must impair extrapolation.

The result extends beyond point error without yielding a universal story of misplaced confidence. At extreme dilution, research produces lower empirical coverage than no research for all three nominal 80% intervals, while the no-research intervals are substantially wider. An interval score penalizing both width and misses favours research for pH, favours no research for dissociation fraction and is similar for precipitation signal. Across all queries, research improves interval

scores for all three responses. A conservative sensitivity analysis allowing the missing session's macro error to take any value in its legal range does not reverse either the overall benefit or the dilute deterioration (Methods).

Crystallization exposes a complementary pattern in the use of evidence already acquired (Fig. 6c; Extended Data Fig. 3). For each of 42 valid forecasts, transparent references use only that same session's source observations. Five-world-equal purity error is 0.03853 for the agent and 0.00510 for the overall source mean, which is more accurate in every valid session. Recovery shows the opposite pattern: agent error is 0.10129 against 0.20443, with the agent outperforming the mean in 39 of 42 sessions. Conditioning on the exact committed solvent set preserves both aggregate directions. Size and fines yield 5/42 and 23/42 wins, respectively. These references diagnose how acquired evidence is used across responses; they are neither no-research controls nor alternative autonomous researchers, and do not estimate the causal benefit of crystallization research.

The new matched-world investigations provide a complementary account of useful understanding with unresolved mechanisms. Across their twelve sealed queries, world-equal errors are 0.00196 for normalized pH, 0.00312 for free dissociation fraction and 0.00243 for precipitation. Forecast centres preserve scale and path invariance for pH and free dissociation, while the process records retain uncertainty about the underlying chemical explanation. These results show how empirically developed relationships can support later inference without uniquely identifying a mechanism. The block has no contemporaneous no-research arm, and its errors are not an improvement estimate relative to the earlier, chemically different worlds.

This response-specific comparison complements the equilibrium result. The challenge is not only to anticipate change beyond an observed region, but also to preserve a relationship when it remains useful. Within the same crystallization histories, predictions extract information about varying recovery more effectively than the source mean, yet fail to match that simple reference for stable purity. Successful evidence use in one response does not establish that all acquired relationships constrain subsequent judgments equally. Across objectives, priors, budgets and research access, scientific progress therefore has several observable dimensions: what is achieved

experimentally, what the evidence establishes and how well the resulting relationships guide new judgments.

Discussion

Abduction connects the creative and empirical aspects of autonomous inquiry: an observed pattern can suggest an explanatory possibility, and that possibility can organize a further experiment. The new records make this connection concrete through tests of precipitation responses and dilution, with local relationships retained even when competing mechanisms remain unresolved. They also show why proposing an explanation, obtaining discriminating physical information and establishing an explanation are different accomplishments. Scientific understanding develops through their interaction, rather than being guaranteed by any one of them.

Autonomous scientific inquiry is a process in which agents organize the evidence from which their subsequent judgments form. Across the studies here, research objectives and supplied information enter experimental choices; observations support revised expectations and new tests; and acquired relationships are used with differing success under new interventions. The breadth of the comparisons and the temporal detail of the process records serve complementary purposes. Cross-system results show that experimental achievement and generalization can align or diverge. Continuous records reveal how such outcomes can coexist with substantive scientific activity, including observing a new response, establishing a local improvement and withdrawing an unsupported expectation.

The findings distinguish truth, evidential support and practical use. A forecast can be accurate even when its explanation remains experimentally unresolved; a locally supported relationship can be applied unsuccessfully elsewhere; and a stable empirical finding can fail to constrain later predictions. The Astra and Sol/Opaque histories make the first distinction visible, equilibrium study A makes the second continuous in time, and the crystallization references expose the third within a common evidence history. These differences would be obscured by assigning each campaign a single label of success or failure. Scientific understanding is better examined through particular relationships, their support and their uses than inferred solely from an overall operating or prediction score[12,22,23].

These distinctions also explain why discovery and misunderstanding need not be opposing descriptions of an entire researcher. One investigation can establish a useful relationship and still leave its continuation unresolved. A failed test can produce a warranted revision, whereas a successful recipe need not identify the mechanism that made it work. The relevant scientific question is how the status of a relationship changes as evidence arrives and as the relationship is carried into another decision. Experimental design and strong inference provide principles for selecting discriminating tests[13–15]; the present records allow their implementation, limitations and alternatives in autonomous inquiry to be examined rather than presumed.

The controlled comparisons address different parts of this process. Matched-world replay measures how much the same executed experiments distinguish two specified physical networks; it does not predict which experiments an agent would choose after alternative feedback or prove unique mechanistic identification. The matched opportunity study estimates the contribution of the full research package under a common assessment protocol, finding overall predictive gains together with a regional reversal in the tested worlds. It does not isolate autonomy from feedback, supplied observations or recording tools. A separate fixed-recipe quench intervention identifies physical effects of an operation without providing outcomes to an adapting agent (Extended Data Fig. 4). The process records establish temporal connections between public judgments and actions, but do not independently identify the effect of a particular judgment or a self-confirming sampling loop. Further interventions can distinguish those explanations without making a successful repair strategy a prerequisite for studying how knowledge forms.

The study is limited to selected programmable systems, a small fixed set of worlds and, for repeated process analyses, one model configuration. Fresh sessions are not controlled model-seed replications or independent samples of world laws. Notebook statements are public records rather than access to private cognition; gaps restrict temporal inference. Cases are retrospectively selected from systematically retained records, with single-analyst interpretation and no second-rater reliability estimate. Historical and current protocols retain separate provenance, including the corrected crystallization block and all failures. These limits constrain claims about prevalence and generality while leaving the observed transitions and bounded comparisons available for scrutiny.

ChemWorld brings the organization of inquiry and its epistemic consequences into the same empirical view. It connects a researcher's choices to the evidence obtained, the relationships proposed and the decisions those relationships subsequently inform. This perspective extends evaluation beyond the completion of experimental cycles to the scientific content produced through them. The central question is how autonomous exploration develops reliable, transferable understanding, and how that developing understanding opens further inquiry. By making these interactions observable and testable, ChemWorld provides an experimental basis for studying the transition from autonomy in research actions to autonomy in scientific exploration and knowledge creation.

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Methods

Worlds, interfaces and experimental units

ChemWorld compiles compatible process components into persistent virtual chemical environments. The public contract specifies materials, legal operations, instruments, resources and task objectives. Private parameters determine each world's responses. Agents access the laboratory and public records, without evaluator state, simulator code, private parameters or withheld reference outcomes. Operations change samples; measurements may consume material. Candidate transitions are validated before commitment. Rejected operations do not constitute the requested physical intervention, although attempts can consume contract-defined allowances. Exact replay checks environment transitions, not regeneration of model decisions.

The broad study covers electrochemistry, partitioning, reaction and thermal processing, aqueous equilibrium, reaction-coupled crystallization and purification. These are controlled virtual systems, not experimentally validated predictions for named physical substances. Current repeated studies use equilibrium and crystallization. Five fixed worlds recur across arms and repetitions; their parameters, initial states and keyed observation mappings are fixed. Repetitions are fresh model sessions, not independent world draws or controlled model-seed replications. The model interface does not expose a usable sampling seed. Repeated identical interventions can share keyed observations, so they are not independent noise samples.

Evidence layers and retained denominators

The historical broad pool contains 240 Sol-medium campaigns and 60 additional equilibrium campaigns from Luna, Terra, GPT-5.5 and Astra, requested at medium reasoning effort. The five-model equilibrium comparison includes fifteen Sol campaigns already in the 240. Requested model labels do not establish equal realized computation. The original 240 include 60 electrochemical, 30 partitioning, 60 reaction, 45 equilibrium, 30 crystallization and 15 purification campaigns. They retain 3,597/3,600 final assays, all 720 assessment stages and 238 protocol-conforming campaigns under their original rules. The additional sixty retain 720 batches and

240 stages. Two original source-completion failures remain included. Earlier selected notebook studies comprise another 24 sessions and are not pooled into the broad or current process cohort.

The current process cohort combines the 45 equilibrium sessions from the earlier 48-session block and its 42-session extension with a complete replacement block of 45 crystallization sessions under a consistent recovery description. Each system covers five worlds, Opaque/Aligned/MisIndexed information and three sessions per cell, with twelve research batches per session. Equilibrium retains 39 complete chains and 41 valid forecasts; corrected crystallization retains 41 and 42, respectively. The combined current denominator is 90 sessions, 80 complete chains and 83 valid forecasts. Old crystallization sessions remain protocol-affected historical evidence. They are not added to the replacement sessions when calculating current outcomes or process frequencies.

Corrected crystallization has 510/540 source batches, 125/135 assessment stages, 145 notebook versions and 42 recommendation retests. All 45 source trajectories and 42 retests pass replay. Four incomplete chains remain: three lack forecasts, and one retains a valid forecast but lacks the subsequent retrospective. The equilibrium subset and the separate opportunity block retain their own failure tables. Missing outcomes are not set to zero, and later favourable outcomes do not replace completed results. Evidence integration does not change the original development or qualification status of any block.

Information, autonomy and recording

Historical goal comparisons pair independent discovery and optimization campaigns within world, information arm, batch budget and prior locus. Electrochemistry contributes thirty pairs across twelve- and twenty-four-batch conditions; reaction processing contributes thirty across parameter- and structural-prior blocks at twelve batches. Operating outcomes are independent retests of sealed recommendations. The joint-outcome counts use public-score prediction MAE, whereas the reaction prior comparison uses the mean across its six normalized response errors. These outcomes are not interchanged. Budget comparisons likewise concern independent sessions with complete resource envelopes; they do not estimate the effect of continuing the same investigation.

The historical five-model equilibrium comparison contains five worlds, three information arms and one twelve-batch campaign per cell: seventy-five campaigns, including the fifteen Sol campaigns already counted in the broad study. The 1 mM threshold descriptively summarizes the lowest positively loaded nominal concentration assayed; it is not a discovery criterion. Concentration is based on nominal recipe loading, and the three dilute assessment queries are shared across configurations. Forecast errors, original prediction intervals and acquired evidence are retained for all arms. The selected Astra and electrochemical trajectories are retrospective process illustrations, with scientific accounts written after experimentation; they are not contemporaneous notebook records or additional replications. Complete historical design, response tables and case values remain in Supplementary Information.

Opaque supplies the public task and operational context without the additional instance dossier. Aligned supplies partial information appropriate to the system. In equilibrium, this includes an effective-acidity interval and an anchored local dilution relationship. MisIndexed uses another world's archival relationship with the same format and qualification. In crystallization, Aligned gives nominal descriptors for anonymous materials; MisIndexed swaps solvent S1/S3 descriptors while preserving S0, S2 and catalyst descriptors. Neither dossier specifies the complete world law.

The original equilibrium Opaque/Aligned sessions precede its MisIndexed extension. Corrected crystallization is a later complete block. These descriptive comparisons are not all concurrently randomized treatments, and new versus old crystallization is not an estimate of the causal effect of correcting the prompt. The new opportunity comparison uses a common contemporaneous protocol, described below.

Agents choose operations, quantities, measurements, repetitions and grouping within fixed budgets. Equilibrium permits twelve source batches, 180 operation attempts, twelve intermediate instruments and twelve final assays; crystallization uses its fixed twelve-batch materials envelope and 720-attempt ceiling. A notebook role asks for an initial testable expectation when available; subsequent updates are agent-selected, without a write-every-batch requirement or automatic reinjection. The role may influence research and is not assumed passive. Notes retain version and public-event cursor. These public records do not expose private cognition. Ab-

sence of a notebook-read tool call does not establish absence of access to the conversation's earlier content.

Assessments and recovery definition

After source research and any operating recommendation, agents seal a scientific account, answer twelve fixed new-intervention queries without laboratory tools and then reflect without truth feedback. Equilibrium includes a structured supplemental assessment. The evidence notation K1/Q/K2/EQS denotes these stages. Queries are hidden during research; post-assessment proposals are not treated as executed source experiments.

Equilibrium errors compare predictions with the mean of five reference observations per query. Coverage checks all five observations. Crystallization uses the noiseless full-process response immediately before final-assay sampling; source observations are noisy. The corrected block retains sixty reference batches, numerically matching the earlier reference outcomes, and separately retests each available committed recommendation.

The corrected recovery definition is seed-excluded recovered solid target divided by the tracked initial reagent charge, in moles, clipped to [0,1]. Source instructions, prediction prompts, observations and scores now refer to this same full-process quantity. The internal crystallization-stage metric instead divides by target product before separation. Earlier process instructions described this latter denominator while observations and scoring used initial reagent charge. Original scores are preserved as protocol-affected diagnostics; they do not enter the corrected cohort. The new block was rerun in full rather than retrospectively relabelling those scores.

Quality requires particles, purity at least 0.80 and fines at most 0.50; joint recommendation success also requires corrected recovery at least 0.10. Corrected joint counts are 13/15 Opaque, 10/14 Aligned and 10/13 MisIndexed among available retests, against fifteen planned sessions in each arm. Particle size is the public index min(d50/250 micrometres, 1); its error is not measured in micrometres.

Research paths and process interpretation

We examine abductive inquiry through publicly stated candidate explanations and relevant subsequent tests, distinguishing these from deductive expectations, empirical support and later application. The focal equilibrium interpretation connects an initial expectation, discrepant observations, a proposed alternative, its anticipated consequences and an executed test. We separately inspect whether the implemented intervention can discriminate the proposed alternatives. Scientific accounts and accurate forecasts alone do not establish this sequence. These records support a case-based process interpretation, not a validated scale of abductive ability or an estimate of mechanism prevalence. The 92 action-linked windows describe recording coverage in the current process cohort; they do not count successful abductions or contain the focal opportunity-study case.

For the retrospective explanation-history panel, selection is fixed to the focal equilibrium opportunity session and the four corrected-crystallization cases already illustrated in the preceding manuscript: lower charge, interpolation, and the two contrasting seed accounts. We trace seven related judgments across these five sessions, retaining all notebook versions, including terminal revisions. Fields distinguish the phenomenon, a candidate or expectation actually stated by the agent, first appearance, conditional prediction, executed operations, observations, subsequent standing and later use. Missing conditional statements are left missing rather than supplied by the analyst. The equilibrium loading expectation, component hypothesis and background interpretation are connected strands of one inquiry, not three independent discoveries. The panel is not sampled to estimate frequencies; no explanation-level prevalence or reliability denominator is inferred from the window counts.

The focal operation-semantics check uses the retained execution version and trajectory's mechanism identity, not current runtime assumptions. That world's single non-catalytic feed is selected both by the reagent operation and by component 0; both add to the same initial-amount ledger read by the equilibrium response calculation. Switching these commands at equal loading therefore does not switch chemical identity. Final nominal concentrations in Extended Data Figure 5 divide committed added moles by total added solvent volume; the staged-dilution batch is plot-

ted at its final 0.8 mol/L. No new simulator calls are required for this inspection. Proposed low-loading and orthogonal measurements remain unexecuted in the retained source history. The scope of the resulting plateau is assessed independently of the agent's proposed mechanism.

Retrospective path extraction covers sixty historical electrochemical, forty-five equilibrium and forty-five corrected crystallization sessions. Fifty within-world condition contrasts retain goal, budget, information and cohort. Counts use committed operations; solvent coverage includes additions during resuspension. Rejected attempts do not establish successful material use. Interrupted sessions remain in the planned denominator.

Structured extraction considers all 178 adjacent pairs among 268 notebook versions in the current 90 sessions. Eligibility requires an earlier note, at least one new committed final assay before the next note, and a committed research operation after that later note; measurement and termination alone do not satisfy continuation. All 178 pairs contain new assays. Ninety-two have the required continuation: 34 equilibrium and 58 crystallization windows in 30 and 35 sessions, respectively. The other 86 pairs (44 equilibrium, 42 crystallization) have no later committed research action and remain available in the evidence archive. The 25 sessions without a complete window are not classified as having learned nothing.

Eligibility is computed from event timing before semantic plan matching, without testing whether a proposed action succeeds. Nevertheless, the rule conditions the sample on continued committed research and excludes terminal revisions from this particular action-linked denominator. All 92 eligible windows are coded. Labels concern one selected immediate plan, not every proposal in a note: 83 matched, seven partly matched and two without an explicit next plan; corrected crystallization contributes 52 matched and six partly matched. Full notes retain abandoned, rejected and altered proposals even when another selected action matches. All 182 update/plan quotations and their assay/action cursors are verified against source records. The archive includes every adjacent pair, selected code, full note and execution history.

The same analyst performs coding and internal scientific review. There is no independent second-rater agreement estimate, blinding or claim of adjudicated mechanism prevalence. The team has assigned review to this workflow; completed internal review is distinguished from independent replication. Counterexamples include warranted retractions, correct recognition of

unexecuted tests and accurate but explicitly conjectural extrapolation. Selection of illustrative cases is retrospective.

Matched chemical worlds and explanatory histories

The additional prospective block contains five parameter pairs, two chemical networks and three fresh Sol-medium sessions per world: thirty planned sessions. Network A couples weak-acid dissociation to direct free-ion precipitation; B adds neutral aqueous ion pairing. Within each pair, acid dissociation, solubility-product, cation, activity and initial-state parameters are matched, along with the public operational contract. Two actual starter assays, each using 6×10^{-7} mol in 0.080 or 0.075 L, are supplied before autonomous research. The paired noise-free responses differ by no more than one declared final-assay standard deviation in each channel. Participants receive chemical framing and loading, dilution and scale dimensions, but not the private network identity or parameters. The initial notebook therefore follows supplied observations, not an evidence-free pretest.

Qualification of the implemented candidate covers 230 physical states and assays, thirty exact replays and 120 runtime mean checks. Five bounded nuisance refits do not adequately fit network B with the tested direct-network family; this check does not prove identifiability against all alternative explanations. An earlier failed physical candidate remains separate. These checks concern the world's implemented information structure, not agent learning.

Each session has twelve autonomous batches followed by a sealed scientific account, twelve query forecasts and a retrospective without truth feedback. The queries are withheld during research, although agents can visit corresponding physical conditions. All thirty sessions complete all stages: 360 source final assays, ninety sealed stages and 1,080 query-response predictions. Eighty notebook versions comprise twenty sessions with three notes and ten with two. Thus continuous midpoint revision cannot be assumed in every session. Three fresh sessions per world share fixed observation keys; the model sampling seed is unavailable.

Every committed source action sequence is replayed in its own and the paired world: thirty source replays and sixty own/partner trajectories pass. Separation is the absolute difference between noise-free response means divided by the final-assay standard deviation: 0.002 for

normalized pH and 0.006 for each fraction. Of 360 final assays, 326 separate at least one channel by three standard deviations and 158 separate at least two; all thirty sessions include an assay of the latter kind. This threshold describes physical information, not statistical significance, intended discrimination or successful abduction. Counterfactual replay fixes actions and does not estimate how an agent would adapt to different observations.

Forecast errors use the mean of five retained reference observations per query; intervals are evaluated against all five observations, which are shared across sessions in the same world. Session values are averaged within each world, then the ten worlds equally. Intervals are nominal 80% prediction intervals, not confidence intervals for means. Their high empirical coverage must be read alongside their widths and does not by itself establish calibration. This block has no no-research arm and uses different chemical laws from the opportunity comparison.

All eighty notes and ninety sealed stages are read with their executed actions. Sixty-seven selected focal histories retain 365 verified quotations. A separate retrospective reading covers 105 existing sessions and 309 notes, tracing 112 selected histories and 218 note anchors; four selected sealed follow-ups add thirteen anchors. The 105 are the current ninety plus fifteen research-opportunity sessions, not additional experiments. Histories can overlap within a session and include terminal revisions. They are descriptive units selected for explanatory relevance, not independent discoveries or a population rate of abduction. Initial expectations, candidate explanations, empirical relations, executed tests and subsequent uses remain distinguishable; absent causal or conditional statements are not supplied by the analyst.

Source operations retain 1,901 attempts: 1,893 committed, six rolled back and two rejected by validation. Another 131 recovered tool failures are counted separately. The first launch attempt contains four startup failures, twenty-six unstarted sessions and no scientific operations; after a thread configuration repair, the entire thirty-session block restarted. All earlier failures and resource records are retained. No completed scientific outcome is replaced because of its value.

Source-only forecast references

Two transparent references are computed for each valid corrected-crystallization forecast using only that researcher’s successful source final assays. The first averages all available assays.

The second averages assays whose exact set of committed solvent identifiers matches the query, falling back to the overall mean when none match. Solvent sets include successful solvent additions and resuspension, excluding rejected actions. The fitting function receives no test truth. These references are descriptive calculations, not separately trained agents or mechanistically complete baselines.

There are 42 valid sessions and 504 queries. Same-solvent fallback occurs in 186/504 queries: 1/180 Opaque, 95/168 Aligned and 90/156 MisIndexed. Matching does not control thermal history, quantities or final liquid composition. References have no predictive intervals. Each response is reported in its own public scale; no combined score is formed across recovery, purity, size and fines.

The added distribution analysis uses those same 42 valid sessions and their 504 source final assays. For each response we expose source observations, all query reference means, forecasts, signed errors and session summaries. Descriptive pooled means and quantiles retain repeated query targets; they are not independent-condition estimates. Signed error is forecast minus reference mean. The world-equal signed mean first averages twelve queries within each session, then sessions within each world and finally the five worlds. Three missing forecasts remain in the planned denominator of 45. These calculations use only the corrected cohort and do not import purity levels or directional bias from historical crystallization.

Matched research-opportunity comparison

The fixed design contains five equilibrium worlds, Research/NoResearch and three fresh Sol-medium sessions per cell, for thirty planned sessions. All use Opaque information. The shared public background contains the task and measurement definitions, material catalogue, legal operations/instruments and initial state; the instance archival prior is null. Opaque therefore means absence of an additional instance-specific law, not absence of public chemical framing or model pretraining. The complete initial public contexts accompany the manuscript. Fifteen within-world/repeat checks confirm scientific-input equality after removing only the non-scientific campaign identifier. Research additionally receives twelve autonomous source batches and the ex-

isting notebook opportunity; NoResearch receives no source operations or observations. Both arms use identical assessment prompts, schemas, order and restrictions.

All thirty sessions reach terminal status. Research completes 180/180 batches; the combined arms complete 117/120 assessment stages and provide 29 valid forecasts. In world three's third Research repetition, the forecast stream disconnects after the scientific account. Q and the two later stages are missing, without replacement. Thus Research has fourteen valid forecasts, NoResearch fifteen; Research world three has two rather than three. Fifteen source trajectories and twenty-five reference trajectories (300 reference batches) pass exact replay. The retained resource ledger is complete for 29/30 sessions; one has only recorded partial usage. Recorded input plus output totals 13,428,656 tokens. Missing usage is not zero and monetary cost is unavailable.

A first execution attempt failed because the assessment transport configuration omitted a required field. Four launched sessions, two completed source histories totalling twenty-four batches and their resource costs are retained separately; no scientific-account stage completed. After fixing that execution defect, the whole fixed thirty-session block restarted. This was a protocol repair, not replacement based on scientific outcomes. Only the complete restarted block enters the comparison.

The comparison estimates the total research-opportunity package under a shared assessment. It does not isolate autonomy relative to supplied data, feedback alone, notebook use or private representational change. It is between fresh sessions, without a within-session pretest. World and arm order are balanced in a predetermined launch schedule, not a random sample of world laws.

Quantitative summaries and missingness

Equilibrium macro MAE equally weights normalized pH, dissociation fraction and precipitation signal across the designated queries. The extreme-dilution group comprises three queries fixed before the repeated blocks; the remaining nine are not assumed entirely in distribution. Session errors are averaged within each world, then the five worlds receive equal weight. Individual

sessions and all missing stages remain visible. Responses, queries, reference observations and batches are correlated within sessions and worlds; they are not independent researcher samples.

Coverage and mean interval width accompany the proper interval score for the agents' nominal 80% intervals[22]. With lower endpoint l, upper endpoint u, outcome y and alpha = 0.2, the score is (u-l) plus (2/alpha)(l-y) when y is below l, or (2/alpha)(y-u) when y is above u. Lower is better. Aggregates weight worlds equally. Agent intervals are not confidence intervals for our group means.

A post hoc sensitivity analysis assigns the single missing opportunity forecast any macro MAE in [0,1], the legal error range, and recomputes means using the planned three sessions per arm/world and five equal worlds. Research-minus-NoResearch bounds are [-0.098659, -0.031992] overall and [+0.036768, +0.103434] for dilution. These are conservative bounds on missing-outcome influence, not imputation, confidence intervals or population inference. Three repetitions per fixed condition describe variation without establishing a population reliability rate.

Matched quench intervention

A separate retained block applies the eighteen original first-cohort crystallization recommendations in five worlds, with and without a single quench operation: ninety pairs, 180 trajectories. All other action arguments and initial states are matched. Quench is removed at its recorded location or inserted immediately before the first seed/crystallization-cooling operation. No agent adapts or receives these outcomes.

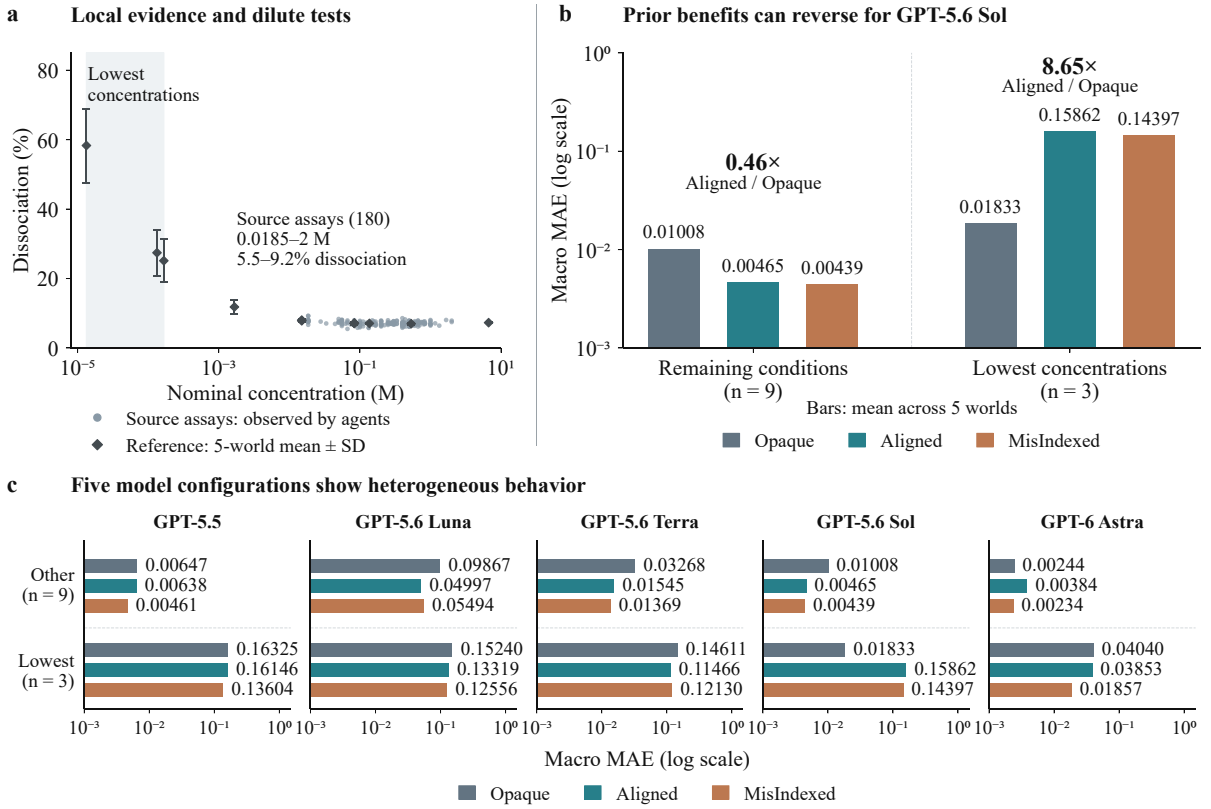
All trajectories are retained and replayed; seventy-five pairs reach both final assays. In five, the absent arm lacks particles; in ten, the present arm makes the subsequent fixed cooling request invalid. Failed arms are not repaired. Among complete pairs, mean present-minus-absent fines changes by +0.30937 and recovery by +0.00741; joint success changes from 34 to 4. These conditional effects describe physical operations under the implemented initial-charge recovery metric. Quench changes multiple physical state variables and does not isolate cooling rate or a cause of scientific judgment. Donor recipes are historical and are not sampled from the corrected block.

Data and code availability

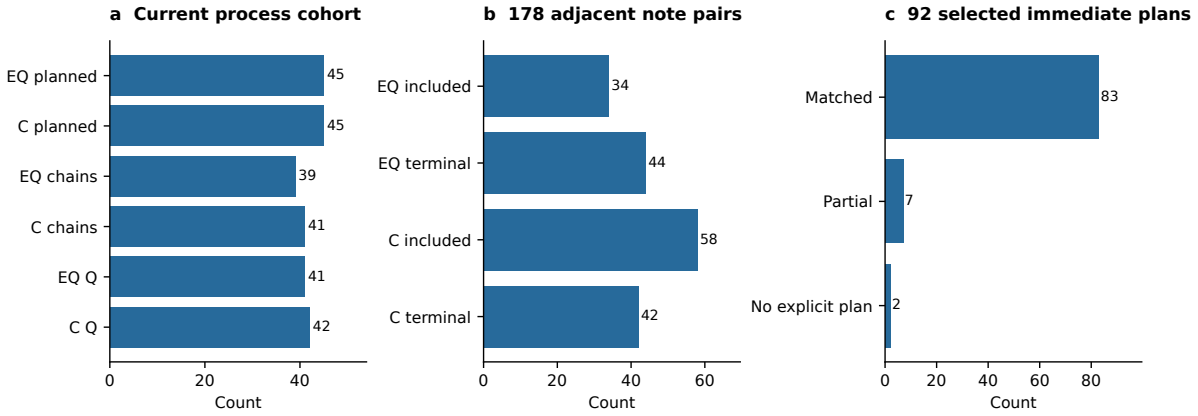
A portable review archive accompanies the manuscript with complete public initial inputs, 1,044 query-response prediction rows, all 270 planned session/group/response cells, world summaries, the continuous focal study, all 178 adjacent note pairs and full public histories for the current 90 process sessions. Missing forecasts remain explicit. Original summaries, exact replay and resource records remain in the project repository; the archive adds no participant evidence or independent second-rater assessment. Supplementary Information reports the numerical decomposition and coding-selection rules. A persistent public deposit remains to be finalized. Manuscript compilation reads retained results without new model or simulator experiments.

Figure 1 is an ImageGen concept illustration. Figures 2–3 and Extended Data Figure 1 retain original native quantitative artwork; all other figures are drawn from retained records and derived summaries. No quantitative point geometry is generated by ImageGen. The accompanying archive adds the selected explanation histories, operation-semantics derivation, focal assay inputs and corrected-crystallization distributions. The archive also contains the thirty matched-world public histories and sealed payloads, quantitative and semantic summaries, selected case anchors and retrospective historical readings. Supporting historical comparisons and separate experimental denominators are detailed in Supplementary Information.

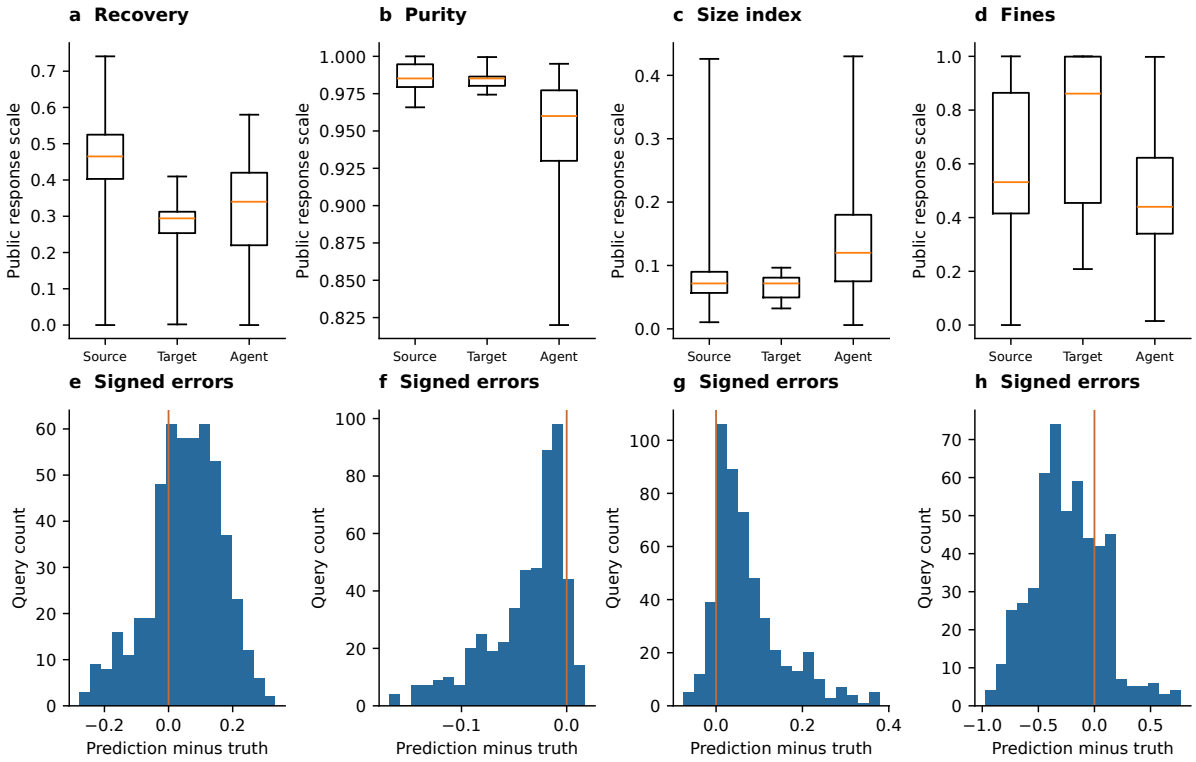
Extended Data



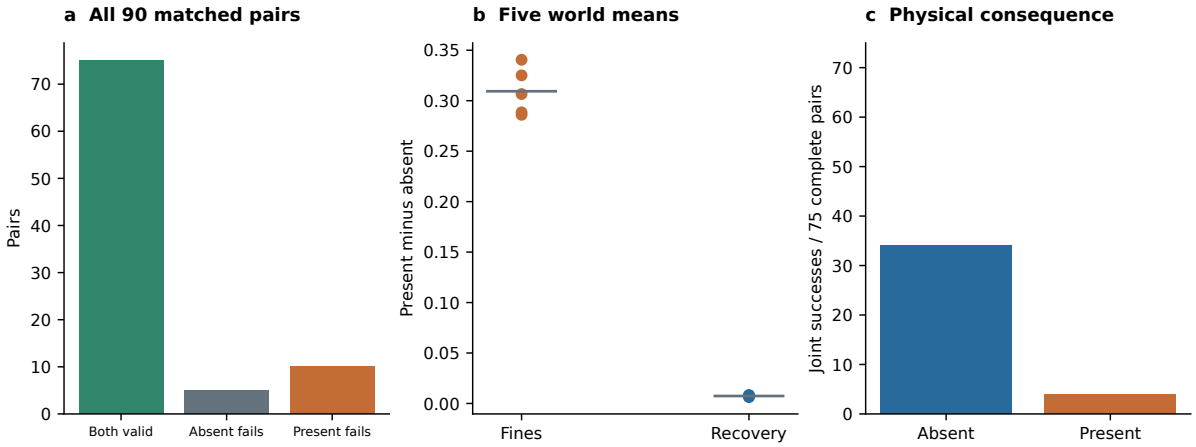
Extended Data Figure 1 | The reach of partial prior knowledge depends on the researcher and the condition. Original equilibrium comparison. a, All 180 Sol source assays and withheld query means across five worlds; error bars are descriptive sample SD, not confidence intervals. b, Five-world macro MAE for Sol under three information conditions. c, All five model configurations on the same logarithmic error scale. Bar baselines are 0.001. The three lowest-concentration queries and remaining nine were a post hoc partition of this historical cohort, fixed before subsequent repeated blocks. These are historical results; the separate repeated-cohort values in the text are not added to these bars. Each model–world–information cell contains one campaign.



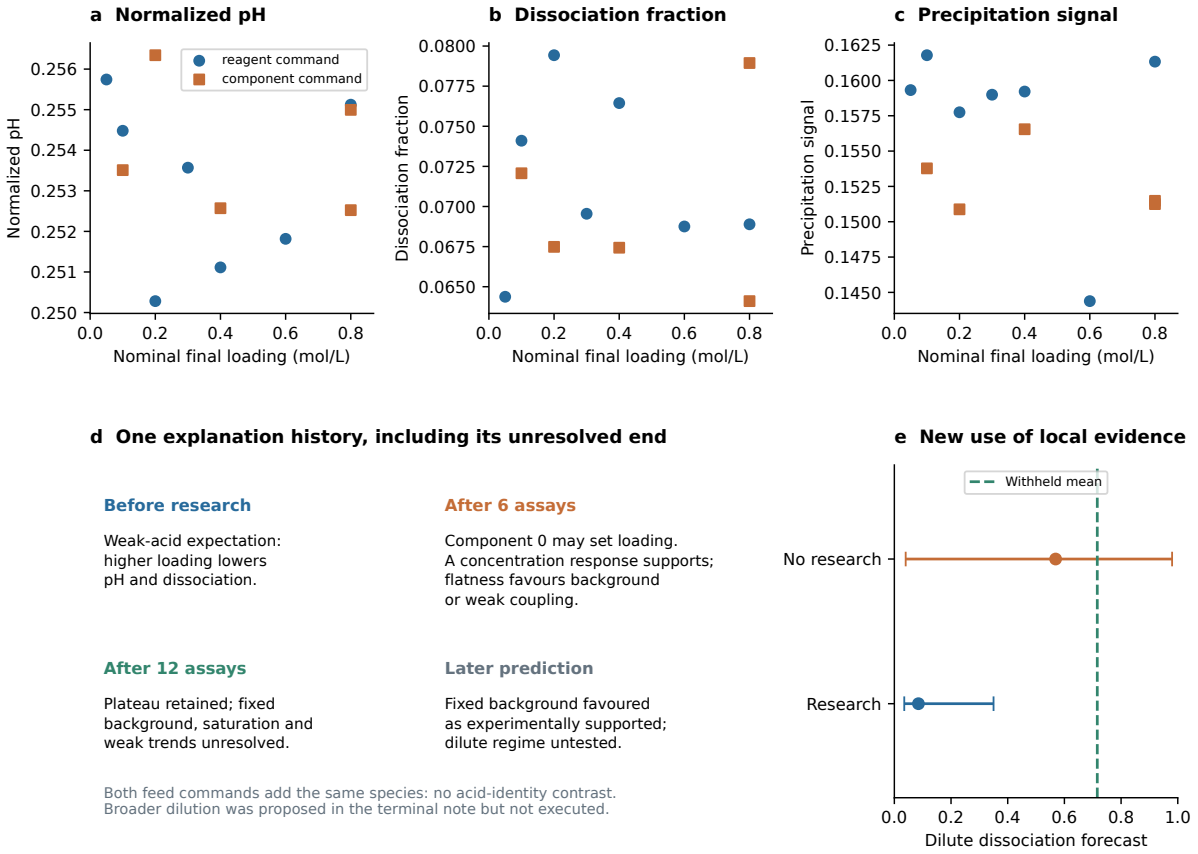
Extended Data Figure 2 | Recording coverage and selected immediate plans. a, Current process-session denominators, complete chains and valid forecasts. b, All 178 adjacent note pairs: 34 equilibrium and 58 crystallization pairs have a committed research action after the later note; the other 44 and 42 do not. The chart's terminal category denotes no subsequent committed research, not absence of revision. c, The 92 selected immediate plans: 83 matched, seven partly matched and two not explicit. These counts describe timing and selected plan–action correspondence, not explanatory hypotheses or learning success.



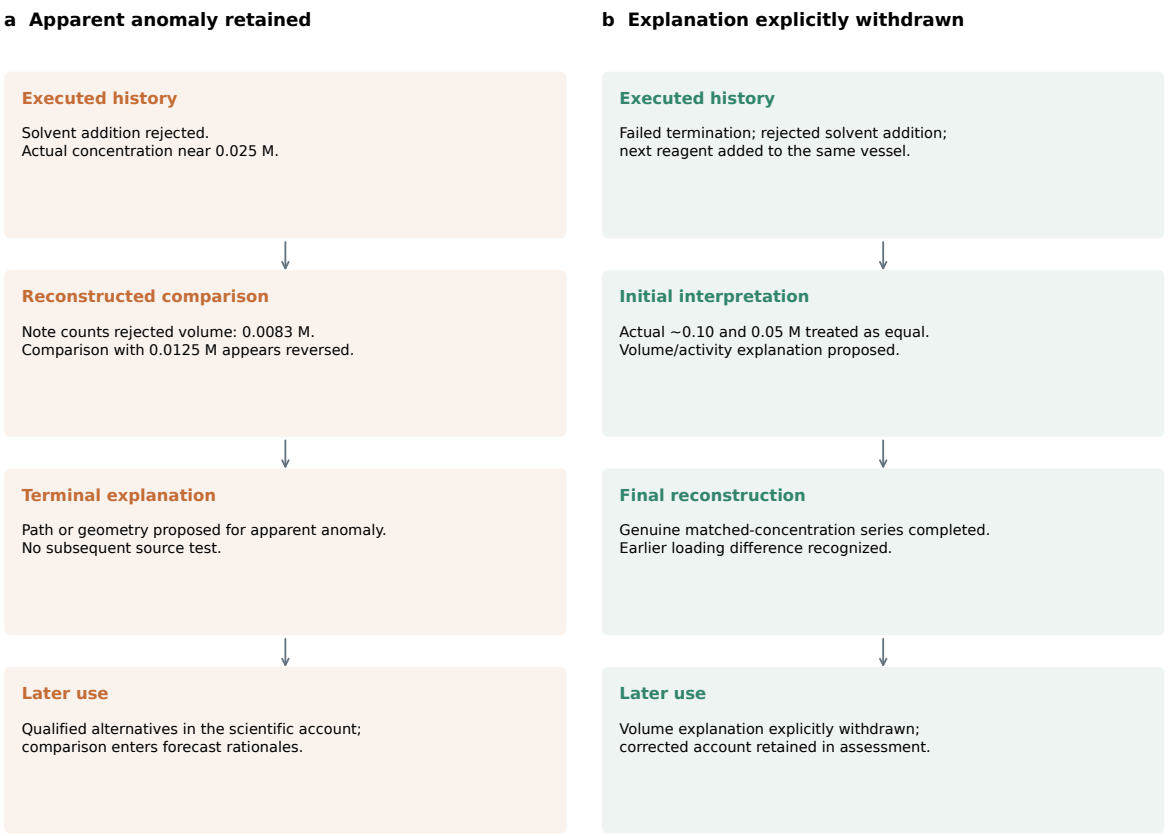
Extended Data Figure 3 | Corrected-crystallization response distributions. a–d, All 504 source final assays from the 42 valid-forecast sessions, their 504 query targets and 504 agent forecasts per response. Boxes show medians and interquartile ranges, whiskers the full range. Target entries repeat across sessions sharing a world; they are not 504 independent conditions. e–h, Signed prediction-minus-reference errors for the same queries. The purity shift is computed from this corrected cohort. Size is the public dimensionless index, not micrometres; response scales differ. Three sessions lacking forecasts remain in the planned denominator of 45 but do not supply values here.



Extended Data Figure 4 | Matched quench effects on physical outcomes. Eighteen historical recommendations are replayed in five worlds with and without one quench operation. a, All ninety pairs and their failures: seventy-five have both final assays, five absent arms lack particles and ten present arms invalidate a later fixed cooling request. b, World-specific mean changes among complete pairs; horizontal marks average five worlds. c, Joint success among the 75 complete pairs. These are physical effects under the implemented initial-charge recovery metric. Agents neither adapt nor receive these control outcomes. No judgment-formation effect is inferred.



Extended Data Figure 5 | The emergence and later use of an explanatory hypothesis. a–c, All twelve final assays in equilibrium study A against actual nominal final loading, calculated from committed added moles and solvent volumes. Circles use the reagent command; squares use the component command. These commands add the same species in this execution version. Batch 11 includes staged dilution to 0.8 mol/L; no fitted curve or uncertainty on a mean is implied. d, The continuous public history connects an initial expectation, an alternative explanation, terminal uncertainty and later use. Text is analytical condensation, not additional agent quotations. e, The same study's dilute forecast and its matched no-research forecast, with the original 80% predictive intervals and withheld mean. All plotted values come from retained records.



Extended Data Figure 6 | Experimental histories acquire and lose explanatory meaning. Two selected matched-world sessions, with operations aligned to public notes and later use. In the first, an uncommitted solvent addition enters the reconstructed concentration and creates an apparent pH reversal. A path or geometry explanation first appears after source research; no further physical test follows. In the second, a purported matched-concentration difference is later recognized as a loading difference, and its volume explanation is explicitly withdrawn. Box contents summarize source records; arrows indicate temporal order, not an isolated causal effect. These are separate sessions, selected retrospectively, rather than randomized correction arms.